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THE UNIVERSITY OF HONG KONG

CIVL2108 Principles of Civil Engineering Management

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Financial Feasibility Study-1

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10 million HKD



Have it today?
10 million HKD

?

Have it in 2023?
10.1 million HKD

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1. Financial Feasibility Study

Have it today?
10 million HKD

Time value of money

1% annum

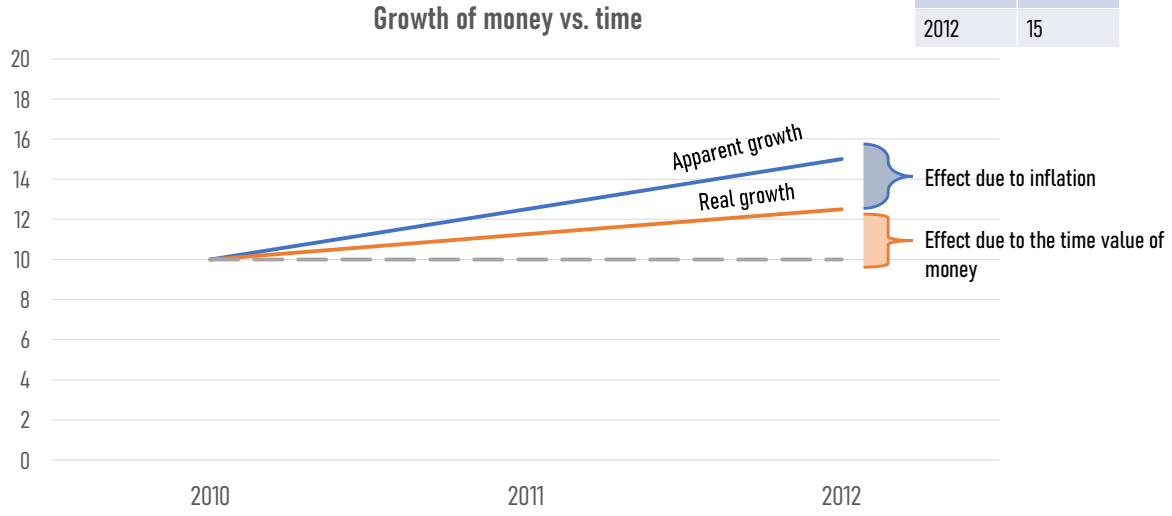
1. **Growth of money: More than 0.1 million HKD can be earned by property investing so that the sum of money will grow and become a larger sum a year later**
2. **Inflation: the purchasing power of 10 million HKD a year later will decline**
3. **Risk: The manager will take all the money and disapear !!!**

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2. The Time Value of Money

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2010	10
2011	12.5
2012	15



THE Value of money in 2012 is lower than that in 2010, i.e., the purchasing power of 15million is less than that of 2010.

3. Mathematics of compound interest

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1. Compound amount factor

Let P = the initial investment of the principal sum

n = the number of periods, which refers usually to months or years

i = interest rate in % per period

S = the total sum of money accumulated after n period

$$S = P(1+i)^n = P \cdot (\text{caf})^{n,i}$$

- $10,000 is put in a fixed account of a bank offering an interest rate of 8% per annum compounded half annually. What will be the compound amount after 5 years?

$S = 10{,}000 \times (\text{caf})^{n=10,\ i=4\%}$
$= 14{,}802$

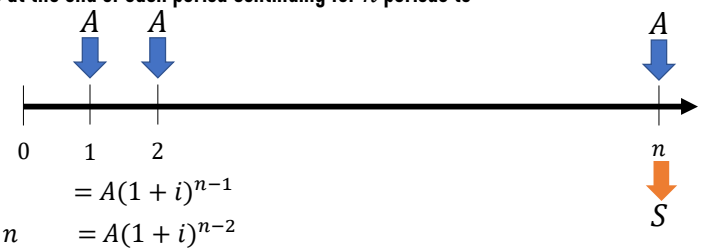


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3. Mathematics of compound interest

2. Uniform series compound amount factor

Let A = the periodic uniform payment made at the end of each period continuing for n periods to accumulate a sum S.



Compound amount of the first A at period n $= A(1+i)^{n-1}$

Compound amount of the second A at period n $= A(1+i)^{n-2}$

Compound amount of the $(n-1)$th A at period n $= A(1+i)$

Compound amount of the nth A at period n $= A$

$= S$

$$S = A + A(1+i) + A(1+i)^2 + \cdots + A(1+i)^{n-1}$$

$$S = A\left[\frac{(1+i)^n - 1}{i}\right] = A \cdot (\text{uscaf})^{n,i}$$

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3. Mathematics of compound interest

- **A manufacturing company will have to replace one of its machines 5 years from now. A new machine costs $125,000. How much is the end-of-year annual instalment the company has to put into a bank in order to save enough money in five years to buy the new machine if the bank is offering an interest rate of 8% per annum?**

$$125{,}000 = A \times (\text{uscaf})^{n=5,\ i=8\%}$$
$$A = 21{,}306$$

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4. Uniform series sinking fund factor

$$S = A\left[\frac{(1+i)^n - 1}{i}\right] = A \cdot (\text{uscaf})^{n,i}$$

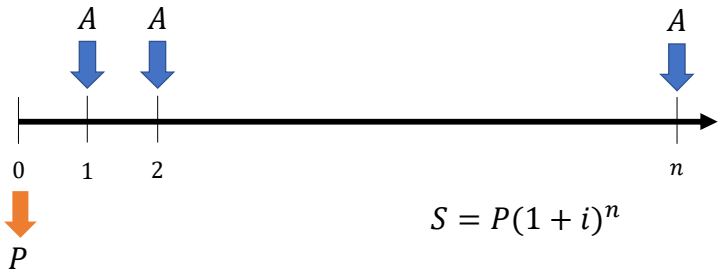
$$A = S\left[\frac{i}{(1+i)^n - 1}\right] = S(\text{ussff})^{n,i}$$

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5. Uniform series capital recovery factor

Initial loan P and subsequent uniform series of payments A to offset against the loan



$$S = P(1+i)^n$$

$$S = A\left[\frac{(1+i)^n - 1}{i}\right]$$

$$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right] = P \cdot (\text{uscrf})^{n,i}$$

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5. Uniform series capital recovery factor

- **A man wishes to hire-purchase a stereo hifi which costs $3,600 The down payment required is $300 and the balance will be paid by 12 end-of-period instalments, once each month. Calculate the amount of each monthly instalment if the interest charged is 24% per annum.**

Money borrowed = 3600-300 = 3300

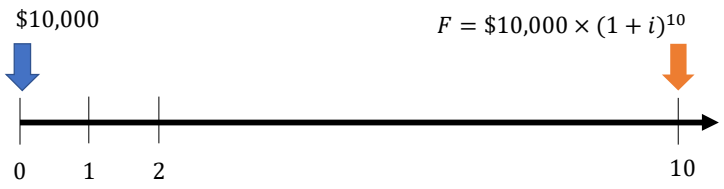
i = 24% p.a. ⟶ i = 2% p.m.

$$A = 3{,}300 \cdot (\text{uscrf})^{n=12, i=2\%} = \$312$$

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6. Present worth factor

Present worth of a single payment



If $i = 10\%$, then $F = 25{,}937$

$25,937 of 10 years from now is equivalent to $10,000 today

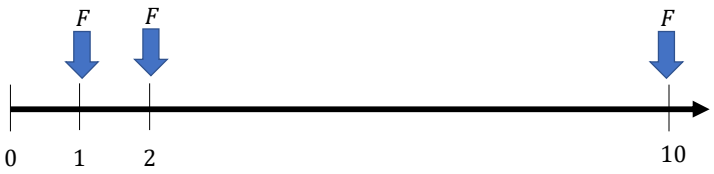
The present worth of $25,937 of ten years later is $10,000

Present worth of a single payment F at nth year

$$= \frac{F}{(1+i)^n} = F \cdot (\text{pwf})^{n,i}$$

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6. Present worth factor



Present worth of n uniform series of payment F

$$= \frac{F}{(1+i)} + \frac{F}{(1+i)^2} + \cdots + \frac{F}{(1+i)^n}$$

$$= F\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$$

$$= F \cdot (\text{uspwf})^{n,i}$$ **Uniform series present worth factor**

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6. Present worth factor

- **A man has borrowed \$100,000 from a bank which charges him 12% interest per annum. He has to recover the loan in 12 months.**

a) **How much should he pay at the end of each month to the bank? (Assume uniform payment)**

$\$10{,}000 \times (\text{uscrf})^{n=12, i=1\%}$
$= \$8885$

b) **If he wants to make an early redemption (i.e. pay all the money which he owes the bank) at the end of month 4, how much should he pay?**

$\$8{,}885 \times (\text{uspwf})^{n=12-\ ,i=1\%}$
$= \$67{,}984$

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6. Present worth factor

The basic idea of compound interest is that the interest is calculated based on the new balance

- Depositing money in a savings account
  $100 $110 $121 $133.1 …

Interest charged to a loan,

Interest will be charged based on the new balance (or the unpaid balance) of the loan

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7. Method 1

The principal sum borrowed ($100,000) be amortized in five years, $20,000 each time

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		20,000	80,000	10,000	30,000
2		20,000	60,000	8,000	28,000
3		20,000	40,000	6,000	26,000
4		20,000	20,000	4,000	24,000
5		20,000	0	2,000	22,000
					130,000

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7. Method 2

The principal sum borrowed be amortized in five years in an increasing manner from $10,000 to $30,000

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		10,000	90,000	10,000	20,000
2		15,000	75,000	9,000	24,000
3		20,000	55,000	7,500	27,000
4		25,000	30,000	5,500	30,000
5		30,000	0	3,000	33,000
					135,000

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7. Present worth

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		20,000	80,000	10,000	30,000
2		20,000	60,000	8,000	28,000
3		20,000	40,000	6,000	26,000
4		20,000	20,000	4,000	24,000
5		20,000	0	2,000	22,000
					130,000

Method 1

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		10,000	90,000	10,000	20,000
2		15,000	75,000	9,000	24,000
3		20,000	55,000	7,500	27,500
4		25,000	30,000	5,500	30,000
5		30,000	0	3,000	33,000
					135,000

30500

Method 2

Method 1: Present worth $= \frac{30,000}{1+0.1} + \frac{28,000}{(1+0.1)^2} + \frac{26,000}{(1+0.1)^3} + \frac{24,000}{(1+0.1)^4} + \frac{22,000}{(1+0.1)^5} = \$100,000$

Method 2: Present worth $= \frac{20,000}{1+0.1} + \frac{24,000}{(1+0.1)^2} + \frac{27,000}{(1+0.1)^3} + \frac{30,000}{(1+0.1)^4} + \frac{33,000}{(1+0.1)^5} = \$100,000$

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7. Method 3

The principal sum borrowed be paid in a lump sum at the end of Year 5

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Unpaid balance (i.e. new balance)	(4) Interest charged to new balance (Col 3 x 10%)	(5) Annual instalment (Col 2 + Col 4)
0	100,000		100,000		
1		0	100,000	10,000	10,000
2		0	100,000	10,000	10,000
3		0	100,000	10,000	10,000
4		0	100,000	10,000	10,000
5		100,000	0	10,000	110,000
					150,000

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7. Method 4

We separate the annual instalment into two components

(1) $P \times i$ = principal sum $\times$ i percent per annum

(2) A sinking fund payment of $P \times (\text{ussff})^{n,i}$

End of year	(1) Principal borrowed	(2) Principal amortization	(3) Annual sinking fund for redemption of $100,000	(4) Interest (same as col 4 in Table 1-3)	(5) Annual instalment (col 3 + col 4)
0	100,000				
1		0	16,379	10,000	26,379
2		0	16,379	10,000	26,379
3		0	16,379	10,000	26,379
4		0	16,379	10,000	26,379
5		100,000	16,379	10,000	26,379

7. Method 5

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- **Annual instalment** $= 100{,}000 \times (\text{uscrf})^{n=5,i=10\%}$

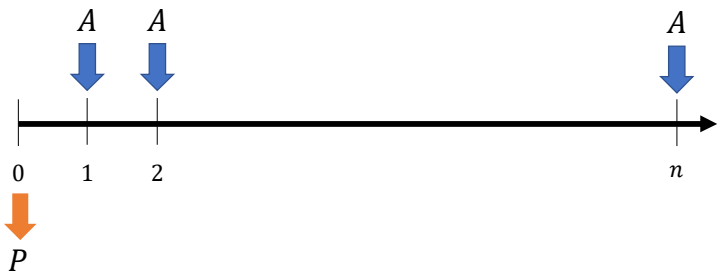
Uniform series capital recovery factor

8. Economic Appraisal of Projects

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- **So far,** i **is the interest rate or the rate of return of an investment**

Generally, it is called discount rate when it is used in the equivalence calculation



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8. Economic Appraisal of Projects

❖ **Equivalent annual cost method**

A man purchases a color TV set which costs him $5000

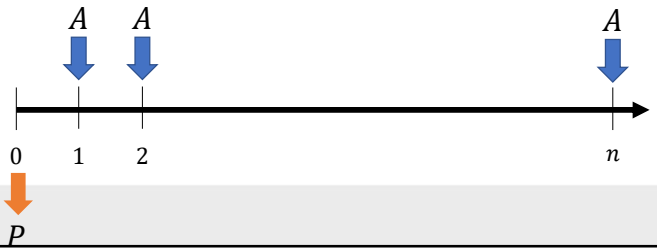
the useful life of it is 5 years, at the end of its life, assuming the remaining value = 0

If $i = 0$

Then , the annual cost to the man of owning it will be $1000

If $i = 10\%$

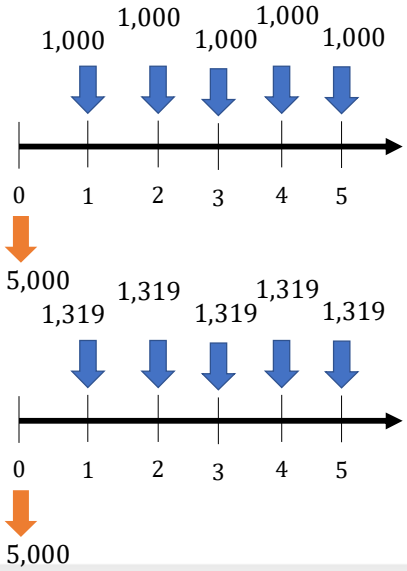
Then , the annual cost to the man of owning it will be $100{,}000 \times (\text{uscrf})^{n=5, i=10\%} = 1{,}319$



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8. Economic Appraisal of Projects



Depreciation of asset at zero discount rate

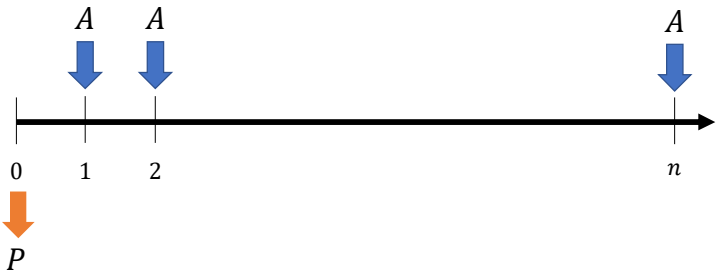
Depreciation of asset at 10% discount rate

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8. Economic Appraisal of Projects

Example 1: The construction cost of a sewerage system is estimated to be $30,000,000. The annual operation, maintenance and repair (i.e., OMR) cost will be $1,000,000 per year. The annual income (i.e. benefit) from the collection of sanitation fees from the users will be $3,500,000. Taking a time horizon of 30 years and a discount rate of 5%, determine if the project is worthwhile. (inflation free assumption)



A: payment
A: annual cost of investment P

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8. Economic Appraisal of Projects

❖ **Equivalent annual cost method**

Annual benefit		3,500,000
Annual OMR cost	1,000,000	2,951,500
Equivalent annual cost of construction	=30,000,000× $(\text{uscrf})^{n=30,i=5\%}$ =1,951,500	
Net annual benefit		548,500

$$\text{B/C ratio} = \frac{3,500,000}{2,951,500}$$

$$= 1.186$$

❖ **Present worth method**

Present worth of benefits in 30 years	=3,500,000 × $(\text{uspwf})^{n=30,i=5\%}$ =53,803,400	53,803,400
Present worth of construction cost	=30,000,000	45,372,400
Present worth of OMR cost	=1,000,000 × $(\text{uspwf})^{n=30,i=5\%}$ =15,372,400	
Net present worth		8,431,000

$$\text{B/C ratio} = \frac{53,803,400}{45,372,400}$$

$$= 1.186$$

Benefit-cost ratio (B/C ratio)

8,431,000 × $(\text{uscrf})^{n=30,i=5\%}$ = 548,500

$$\text{B/C ratio} = \frac{\textit{Equivalent annual total benefits}}{\textit{Equivalent annual total costs}} = \frac{\textit{Present worth of total benefits}}{\textit{Present worth of total costs}}$$

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8. Economic Appraisal of Projects

Example 2: Consider the relative costs of a timber pedestrian bridge and a steel one: their initial capital costs, annual OMR costs and useful lives are given below. Find the alternative of the least overall costs. (use i = 8%) (inflation free assumption)

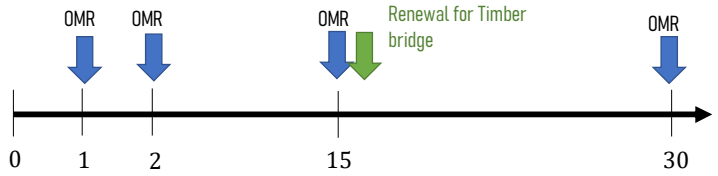
	Timber Bridge	Steel Bridge
Initial capital	$500,000	$700,000
Annual OMR cost	$30,000	$5,000
life	15 years	30 years

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8. Economic Appraisal of Projects

- **Present worth method**

	Timber bridge (15 years life)	Steel bridge (30 years life)
a) Initial capital	$500,000	$700,000
b) Present worth of OMR costs over 30 years	$30{,}000 \times (\text{uspwf})^{n=30, i=8\%}$ = $337,731	$5{,}000 \times (\text{uspwf})^{n=30, i=8\%}$ = $56,288
c) Present worth of renewal	$500{,}000 \times (\text{uspwf})^{n=15, i=8\%}$ = $157,620	Nil
Present worth of total cost	$995,351	$756,288



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8. Economic Appraisal of Projects

- **Equivalent annual cost method**

	Timber bridge (15 years life)	Steel bridge (30 years life)
a) Equivalent annual cost of initial capital	500,000× $(\text{uscrf})^{n=15,i=8\%}$ = \$58,410	700,000× $(\text{uscrf})^{n=30,i=8\%}$ = \$62,174
b) Annual OMR costs	30,000	5,000
Total equivalent annual costs	\$88,410	\$67,174

- **Relations between two methods**

$\$995{,}351 \times (\text{uscrf})^{n=30,i=8\%} = \$88{,}410$

$\$756{,}288 \times (\text{uscrf})^{n=30,i=8\%} = \$67{,}174$

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9. Equivalent annual cost of an asset with salvage value

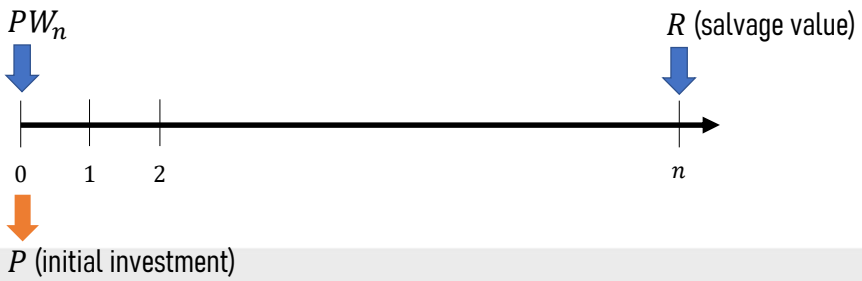
- **Present worth of salvage value**

Let PW_n = present worth of the salvage value R
$= R(\text{pwf})^{n,i}$

∴ net present worth of investment $= P - PW_n = P - R(\text{pwf})^{n,i}$

Let equivalent annual cost of an asset with salvage value $R = A$

$$\therefore A = \left[P - R(\text{pwf})^{n,i}\right] \times (uscrf)^{n,i} = [P - R] \times (uscrf)^{n,i} + Ri$$



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9. Equivalent annual cost of an asset with salvage value

A manager wanted to purchase a piece of new equipment for his company. He had been approached by two agents of different manufacturers offering their newest models. After detailed investigation, he obtained the following estimates. Assuming i =12%, determine for the manager which type of equipment he should purchase.

	Model A	Model B
Cost	$20,000	$15,000
Annual maintenance cost	$1,000	$1,800 p.a.
Salvage value	$2,500	$1,500
Life	5 years	5 years

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9. Equivalent annual cost of an asset with salvage value

- **Equivalent annual costs of the two models**

	Model A	Model B
a) Equivalent annual cost of initial capital	$(20{,}000 - 2{,}500) \times (\text{uscrf})^{n=5, i=12\%} + 2{,}500 \times 12\% = \$5{,}155$	$(15{,}000 - 1{,}500) \times (\text{uscrf})^{n=5, i=12\%} + 1{,}500 \times 12\% = \$3{,}925$
b) Annual maintenance cost	1,000	1,800
Total equivalent annual costs	$6,155	$5,725

- **Present worth method**

	Model A	Model B
a) Initial capital	$20,000	$15,000
b) Present worth of annual maintenance	$1{,}000 \times (\text{uspwf})^{n=5, i=12\%}$ = $3,605	$1{,}800 \times (\text{uspwf})^{n=5, i=12\%}$ = $6,489
c) Less present worth of salvage	$-2{,}500 \times (\text{pwf})^{n=5, i=12\%}$ = −$1,419	$-1{,}500 \times (\text{pwf})^{n=5, i=12\%}$ = −$851
Present worth of total cost	$22,186	$20,638